

The Faddeev–LeVerrier algorithm

The Faddeev–LeVerrier algorithm may be used to determine the inverse, determinant, and characteristic polynomial of an $n \times n$ matrix. The algorithm terminates in n steps, where each step involves a single matrix multiplication and only integer division. It works like magic!

Faddeev-LeVerrier algorithm

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1 given an  $n \times n$  matrix  $A$ 
2  $c_n := 1$ 
3  $N \leftarrow \mathbb{0}$ 
4 for  $k \in (n - 1, ..., 1, 0)$ 
5    $N \leftarrow AN + c_{k+1}\mathbb{I}$ 
6    $c_k := \frac{1}{k-n} \operatorname{tr}(AN)$ 
7 return
8    $A^{-1} = -N/c_0$ 
9    $\det(A) = (-1)^n c_0$ 
10   $\chi(\lambda) = \sum_{k=0}^n c_k \lambda^k$ 
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Also refer to a [Julia implementation](#).

Derivation

Start with the characteristic polynomial of A .

$$\chi(\lambda) = \det(\lambda \mathbb{I} - A) = \sum_{k=0}^n c_k \lambda^k$$

Useful fact.

The *adjunct* of a matrix, $\operatorname{adj}(A)$, satisfies $\det(A)\mathbb{I} = A \operatorname{adj}(A)$.

If A is $n \times n$, then $\det(A)$, and hence the entries of $A \operatorname{adj}(A)$, are degree n polynomials in the entries of A . Hence, the entries of $\operatorname{adj}(A)$ are degree $n - 1$ polynomials.

The entries of $N(\lambda) := \operatorname{adj}(\lambda \mathbb{I} - A)$ are λ -polynomals of order $n - 1$, so $N(\lambda) = \sum_{k=0}^{n-1} N_k \lambda^k$ where N_k are matrices. From $\det(\lambda \mathbb{I} - A)\mathbb{I} = (\lambda \mathbb{I} - A)N(\lambda)$,

$$\begin{aligned} \det(\lambda \mathbb{I} - A)\mathbb{I} &= (\lambda \mathbb{I} - A) \sum_{k=0}^{n-1} N_k \lambda^k \\ &= -AN_0 + \sum_{k=1}^{n-1} (N_{k-1} - AN_k) \lambda^k + N_{n-1} \lambda^n \end{aligned}$$

Equating coefficients of λ with $\chi(\lambda)\mathbb{I}$, we obtain:

$$\begin{aligned} c_0 \mathbb{I} &= -AN_0 \\ c_k \mathbb{I} &= N_{k-1} - AN_k \\ c_n \mathbb{I} &= N_{n-1} \end{aligned}$$

To remember these, just write $c_k \mathbb{I} = N_{k-1} - AN_k$ for all $0 \leq k \leq n$ with the understanding that N_k vanishes outside the range $0 \leq k \leq n - 1$. Equivalently,

$$N_k = AN_{k+1} - c_{k+1} \mathbb{I}$$

gives a descending recurrence relation for N_k in terms of the coefficients c_k .

Finding c_k in terms of A and N_k

This stroke of genius is due to ([Hou, 1998](#)).

Useful fact: Laplace transform of derivative.

$$\begin{aligned} \mathcal{L}\{f'(t)\}(s) &= \int_0^\infty f'(t)e^{-st} \, \mathrm{d}t \\ &= f(t)e^{-st} \Big|_{t=0}^\infty + s \int_0^\infty f(t)e^{-st} \, \mathrm{d}t \\ &= -f(0) + s\mathcal{L}\{f(t)\}(s) \end{aligned}$$

Consider

$$\frac{\mathrm{d}e^{At}}{\mathrm{d}t} = Ae^{At}$$

and perform the Laplace transform to obtain

$$-\mathbb{I} + s\mathcal{L}\{e^{At}\} = A\mathcal{L}\{e^{At}\}$$

and finally take the trace:

$$s \operatorname{tr} \mathcal{L}\{e^{At}\} - n = \operatorname{tr}(A\mathcal{L}\{e^{At}\}) \tag{1}$$

Useful fact: the trace of a matrix exponential in terms of eigenvalues.

If λ_i are the eigenvalues of A then $\operatorname{tr}(A) = \sum_i \lambda_i$. Also, A can be put in Jordan normal form $A = PJP^{-1}$ where J is triangular with $\operatorname{diag}(J) = (\lambda_1, ..., \lambda_n)$. Since it is triangular, $\operatorname{diag}(J^k) = \left(\lambda_1^k, ..., \lambda_n^k\right)$.

Therefore, $\operatorname{tr}(A^k) = \operatorname{tr}(PJ^kP^{-1}) = \operatorname{tr}(J^kP^{-1}P) = \operatorname{tr}(J^k) = \sum_{i=1}^n \lambda_i^k$.

Consequently, $\operatorname{tr}(e^{At}) = \sum_{k=0}^\infty \frac{t^k}{k!} \operatorname{tr}(A^k) = \sum_{k=0}^\infty \frac{t^k}{k!} \sum_{i=1}^n \lambda_i^k = \sum_{i=1}^n e^{\lambda_i t}$.

We now compute the terms in [eq. 1](#).

$$\mathcal{L}\{e^{At}\} = \int_0^\infty e^{(A-s\mathbb{I})t} \, \mathrm{d}t = (A - s\mathbb{I})^{-1} e^{(A-s\mathbb{I})t} \Big|_{t=0}^\infty = (s\mathbb{I} - A)^{-1}$$

Caution.

I'm uncomfortable with these integrals. Why should $\lim_{t \rightarrow \infty} e^{(A-s\mathbb{I})t}$ converge?

Note that from $\chi(\lambda) = \det(\lambda \mathbb{I} - A) = (\lambda \mathbb{I} - A)N(\lambda)$ we have

$$(\lambda \mathbb{I} - A)^{-1} = \frac{N(\lambda)}{\chi(\lambda)} \tag{2}$$

Let $(\lambda_1, ..., \lambda_n)$ be the eigenvalues of A . Then $A - s\mathbb{I}$ has eigenvalues $\lambda_i - s$.

$$\operatorname{tr} \mathcal{L}\{e^{At}\} = \int_0^\infty \operatorname{tr} e^{(A-s\mathbb{I})t} \, \mathrm{d}t = \sum_{i=1}^n \int_0^\infty e^{(\lambda_i-s)t} \, \mathrm{d}t = \sum_{i=1}^n \frac{1}{s - \lambda_i}$$

Recall that the roots of the characteristic polynomial of A are its eigenvalues, so $\chi(s) = \prod_{i=1}^n (s - \lambda_i)$.

$$\operatorname{tr} \mathcal{L}\{e^{At}\} = \sum_{i=1}^n \frac{\mathrm{d}}{\mathrm{d}s} \ln(s - \lambda_i) = \frac{\mathrm{d}}{\mathrm{d}s} \ln \left(\prod_{i=1}^n (s - \lambda_i) \right) = \frac{\mathrm{d}}{\mathrm{d}s} \ln \chi(s) = \frac{\chi'(s)}{\chi(s)} \tag{3}$$

Substituting [eq. 2](#) and [eq. 3](#) into [eq. 1](#), we have

$$\begin{aligned} s\chi'(s) - n\chi(s) &= \operatorname{tr}(AN(\lambda)) \\ \sum_{k=0}^n (k - n)c_k s^k &= \sum_{k=0}^{n-1} \operatorname{tr}(AN_k) s^k \end{aligned}$$

which, expanding and equating powers of λ ,

$$c_k = \frac{\operatorname{tr}(AN_k)}{k - n}$$

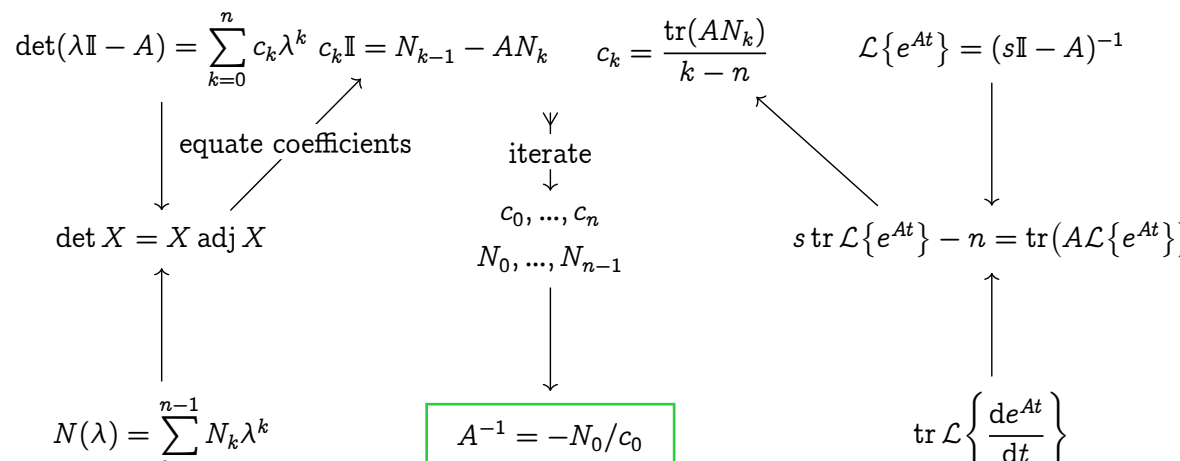
for all $0 \leq k \leq n$ where we define $N_n = 0$.

Useful fact.

The characteristic polynomial coefficients give the determinant, trace, and inverse of the matrix according to:

$$\begin{aligned} \chi(\lambda) &= c_0 + \cdots + c_{n-1} \lambda^{n-1} + c_n \lambda^n \\ &= \det(-A) + \cdots + \operatorname{tr}(-A) \lambda^{n-1} + \lambda^n \\ c_0 &= (-1)^n \det(A), \quad c_{n-1} = -\operatorname{tr}(A), \quad c_n = 1 \end{aligned}$$

Visual summary



References

Hou, S.-H. (1998). Classroom Note: A Simple Proof of the Leverrier–Faddeev Characteristic Polynomial Algorithm. *SIAM Rev.*, 40(3), 706–709. <https://doi.org/10.1137/S003614459732076X>